

Setup Manual

React + Tailwind frontend, Django REST backend, Oracle SQL database

1. What this project is

A professional website for a freelance full-stack web developer. It has two parts that run side by side:

Part	Folder	Technology	Local address
Website (what visitors see)	frontend/	React (Vite), Tailwind CSS	http://localhost:5173
Backend (forms, chat, admin)	backend/	Python, Django, Django REST	http://127.0.0.1:8000
Database	Oracle (or SQLite for a quick start)	Oracle SQL	port 1521

The backend saves each contact-form enquiry to the database first, then emails it. It also runs the optional AI chat assistant, so the API key stays on the server. The site shows clearly labelled **Sample** content until you replace it with your own.

Not included yet: a production web server setup (gunicorn, static files) and automated tests. This manual covers local setup, customisation and a deployment outline.

2. What you need installed

Tool	Version	Check with
Node.js	18 or newer	node -v
Python	3.10 or newer	python --version
Oracle Database	19c or newer (Express Edition 21c works)	optional, see section 6
Oracle SQL Developer	any recent version	optional, to inspect data
A code editors	VS Code and PyCharm	Both

You can skip Oracle at first. With `DB_ENGINE` left empty in `backend/.env`, the backend uses a local SQLite file, so the whole site works without Oracle.

3. Project structure

freelance-site/	
MANUAL.md	this manual
frontend/	
src/siteConfig.js	your name, brand, email, links, tech stack
src/data/projects.js	projects, blog posts, pricing, testimonials
src/data/content.js	services, process steps, "why work with me"
src/pages/	one file per page
src/components/	header, footer, chat widget
public/	CV (PDF), favicon, robots.txt, sitemap.xml
.env.example	copy to .env
backend/	
config/settings.py	all settings, read from .env
enquiries/	enquiry model, form API, AI chat API
enquiries/chat_context.py	what the chat assistant is told
.env.example	copy to .env
requirements.txt	

4. Quick start (about 10 minutes, no Oracle)

4.1 Start the website

Open a VS Code terminal in the project and run:

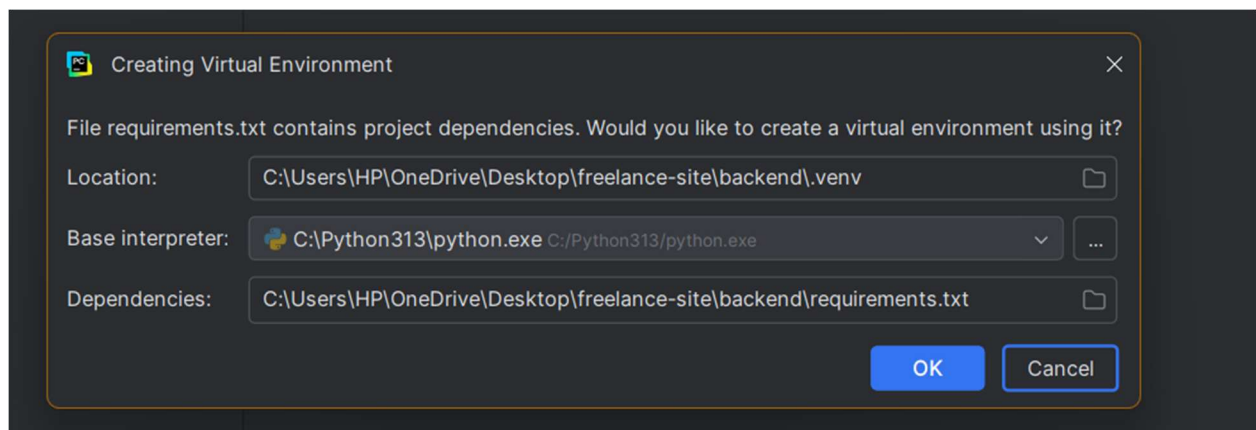
```
cd frontend
npm install
npm run dev
```

Open <http://localhost:5173>. The terminal must be inside the frontend folder. Running npm from the parent folder gives an "ENOENT package.json" error.

4.2 Start the backend

Open the pyCharm terminal in the project:

You should see the pop up window appears, Just Click to OK do not change anything in that window:



4.3 Then in the terminal

```
cd backend  
python -m venv .env
```

Activate the environment:

- Windows PowerShell: `.env\Scripts\Activate.ps1`
- Windows Command Prompt: `.env\Scripts\activate.bat`
- Mac or Linux: `source .env/bin/activate`

If PowerShell blocks the script, run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once and try again. PyCharm can create and activate this environment for you when you open the backend folder.

Then run:

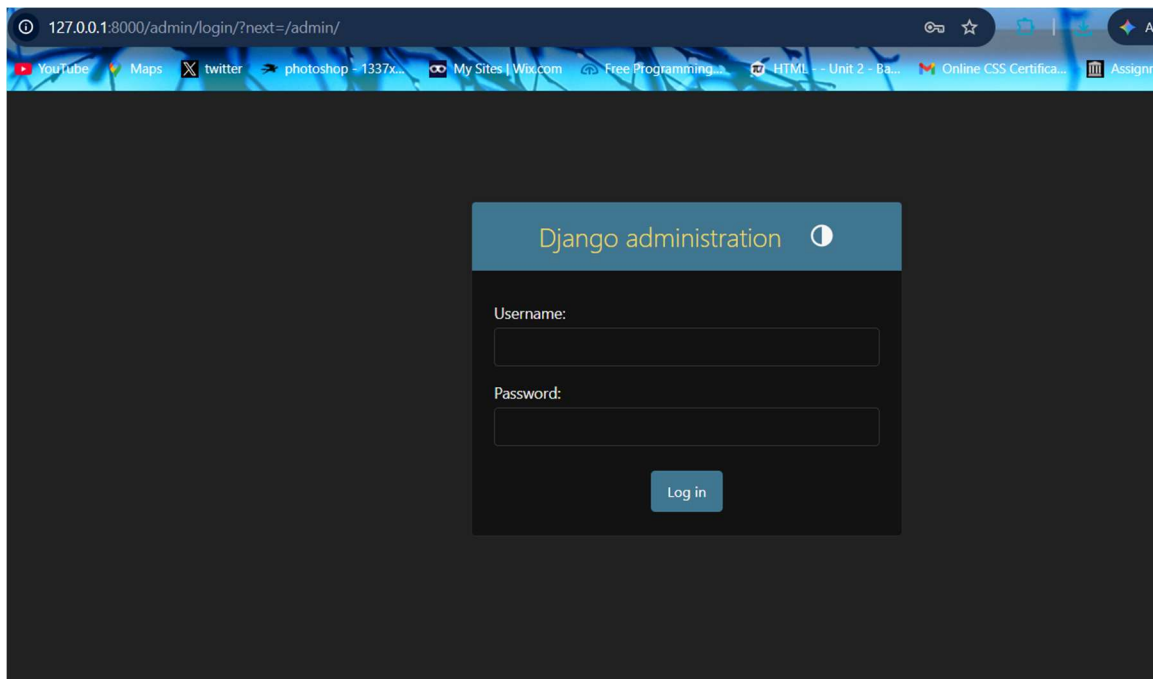
```
pip install -r requirements.txt
```

Windows: `copy .env.example .env` / Mac or Linux: `cp .env.example .env`.

```
python manage.py migrate  
python manage.py createsuperuser  
python manage.py runserver
```

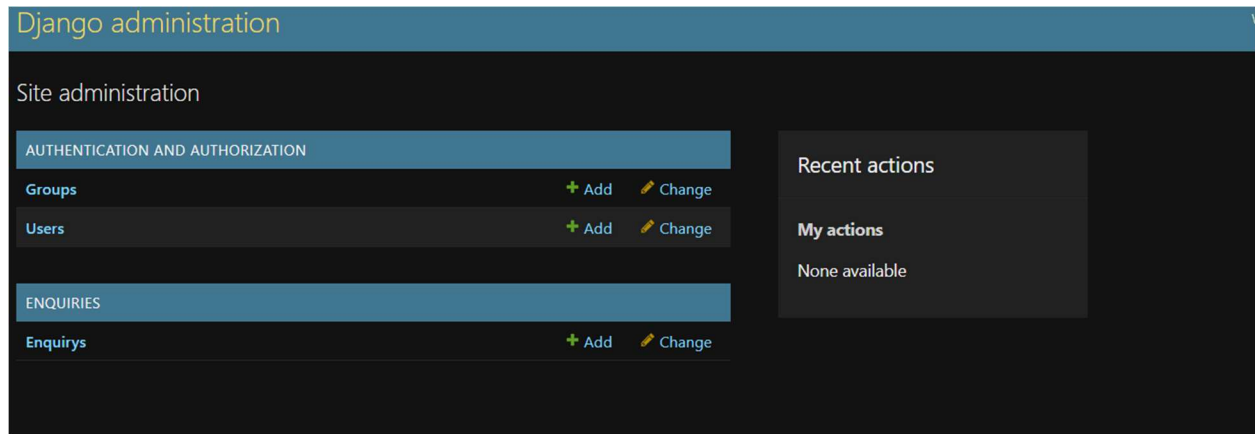
The backend is now at `http://127.0.0.1:8000`. Log in at `http://127.0.0.1:8000/admin` with the superuser you created.

You should be able to see like this :



4.4 Log in by superuser you created:

Then you should see this :



4.5 Connect the website to the backend

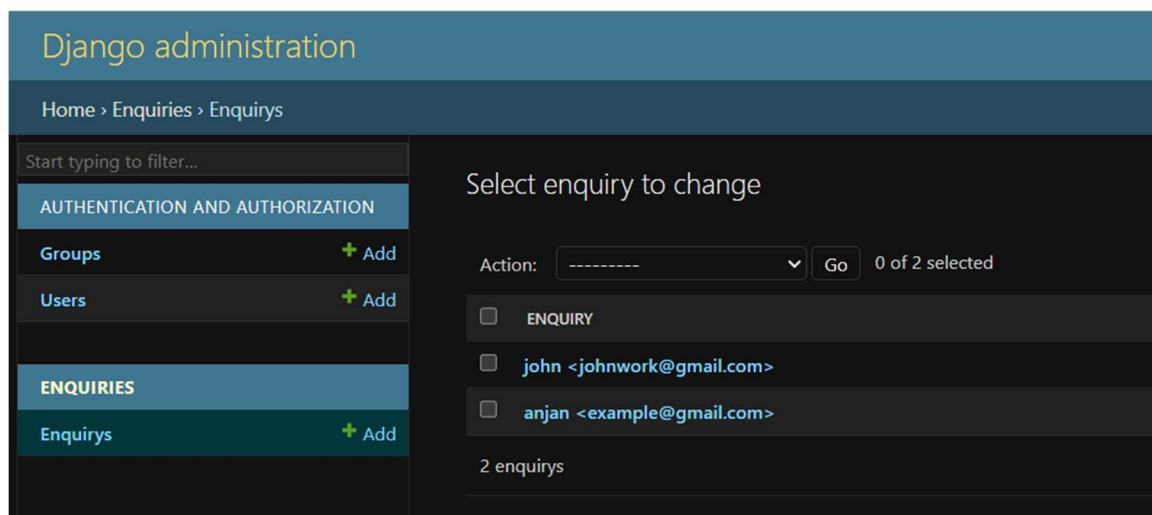
In the frontend folder, copy `.env.example` to `.env`. It contains one line:

```
VITE_API_URL=http://127.0.0.1:8000
```

Stop `npm run dev` with `Ctrl + C` and start it again. Vite reads `.env` only when it starts, and it ignores files named `.env.example`.

4.6 Test it

1. Open `http://localhost:5173/contact`, fill in every field and send.
2. You should see "Enquiry sent."
3. In the admin, open **Enquiries** and click **Change**. Your message is listed.



4. The backend terminal shows the email text, because no real email server is set up yet.

5. The backend settings file

Everything is configured in backend/.env. Never commit this file or share it to git.

Setting	Meaning
DJANGO_SECRET_KEY	Long random text. Change it for any real deployment.
DJANGO_DEBUG	1 for local work, 0 for a live site.
ALLOWED_HOSTS	Comma-separated domains the backend answers to.
CORS_ORIGINS	The website address allowed to call the backend, such as http://localhost:5173.
DB_ENGINE	Empty for SQLite, or oracle.
ORACLE_USER, ORACLE_PASSWORD, ORACLE_DSN	Oracle login, for example localhost:1521/XEPDB1.
EMAIL_*, ENQUIRY_TO_EMAIL	Outgoing email, see section 8.
ANTHROPIC_API_KEY, CHAT_MODEL	AI chat, see section 9.

Write values without quotes or spaces. Avoid # and \$ in passwords.

6. Using Oracle instead of SQLite

6.1 Find your service name

In SQL Developer, right-click your connection and choose **Properties**. Note the host, port and service name. For Oracle XE the service name is usually XEPDB1. You can also run:

```
SELECT sys_context('USERENV', 'SERVICE_NAME') FROM dual;
```

6.2 Create a database user

Connect as SYSTEM to the pluggable database (for XE, XEPDB1, not XE).

Check with SHOW CON_NAME;.

If it says CDB\$ROOT,

run ALTER SESSION SET CONTAINER = XEPDB1; first.

Run each statement on its own, on one line, with **Run Statement** (Ctrl + Enter):

```
CREATE USER freelance IDENTIFIED BY "ChooseAPassword1" DEFAULT TABLESPACE  
USERS QUOTA UNLIMITED ON USERS;
```

```
GRANT CONNECT, RESOURCE TO freelance;
```

Splitting a statement over two lines in the wrong mode makes SQL Developer report “Unknown Command”. If that happens, use `ALTER USER freelance DEFAULT TABLESPACE USERS QUOTA UNLIMITED ON USERS;`.

6.3 Point Django at Oracle

In backend/.env:

```
DB_ENGINE=oracle
ORACLE_USER=freelance
ORACLE_PASSWORD=ChooseAPassword1
ORACLE_DSN=localhost:1521/XEPDB1
```

Then run:

```
python manage.py check --database default
python manage.py migrate
python manage.py createsuperuser
python manage.py runserver
```

The repository already includes the first migration, so you do not need `makemigrations` unless you change `models.py`. Create the superuser again after switching databases, because users are stored per database.

6.4 Look at the data

Connect as freelance and run:

```
SELECT * FROM FREELANCE.ENQUIRIES_ENQUIRY ORDER BY CREATED_AT DESC;
```

If you are connected as another user, add the owner:

`FREELANCE.ENQUIRIES_ENQUIRY`. Without it, Oracle reports ORA-00942.

7. Making the site yours

Everything marked **Sample** must be replaced before the site goes live.

1. **Identity.** Edit `frontend/src/siteConfig.js`: brand, developer name, title, email, WhatsApp number (digits only with country code), booking link (Cal.com or Calendly), and your GitHub and LinkedIn addresses. Empty values hide their buttons.
2. **Projects, blog, pricing, testimonials.** Edit `frontend/src/data/projects.js`. Delete the sample items or set `sample: false`. Add only real projects and genuine testimonials.
3. **Services and process.** Edit `frontend/src/data/content.js`.
4. **About page and FAQ.** Edit `frontend/src/pages/About.jsx` and `Faq.jsx`.
5. **CV.** Replace `frontend/public/My_CV.pdf` with your own file. Keep the file name, or change `cvUrl` in `siteConfig.js`.

6. **Chat assistant.** Edit `backend/enquiries/chat_context.py` so it describes you accurately. It is told never to invent prices, clients or experience.
7. **Colours.** The palette is defined as variables at the top of `frontend/src/index.css`, for light and dark themes.
8. **Search engine files.** After you buy a domain, replace `example.com` in `frontend/index.html`, `public/sitemap.xml` and `public/robots.txt`, and update the structured data brand name in `index.html`.

8. Real email delivery (optional)

Without this, enquiries are saved and the email text only prints in the backend terminal. For Gmail, turn on 2-step verification, create an **App password** in your Google account, and set:

```
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_HOST_USER=yourname@gmail.com
EMAIL_HOST_PASSWORD=your-app-password
ENQUIRY_TO_EMAIL=where-to-receive@example.com
```

Restart the backend. If sending fails, the enquiry is still saved first, so nothing is lost.

9. AI chat assistant (optional)

The floating chat button appears on every page. Add an Anthropic API key to `backend/.env`:

```
ANTHROPIC_API_KEY=your-key
```

Restart the backend. Without a key the chat politely points visitors to the Contact page. The key must stay in the backend only, never in the frontend. Each chat message uses API credit, and chat is limited to 40 requests per hour per visitor. Change the limit in `backend/config/settings.py` (`DEFAULT_THROTTLE_RATES`).

10. Daily routine

1. Make sure Oracle is running, if you use it.
2. Terminal 1, in backend with the environment active: `python manage.py runserver`.
3. Terminal 2, in frontend: `npm run dev`.
4. Open `http://localhost:5173`. Read enquiries at `http://127.0.0.1:8000/admin`.

If you change `backend/enquiries/models.py`, run `python manage.py makemigrations` and then `python manage.py migrate`.

11. Troubleshooting

Symptom	Cause and fix
npm error ENOENT ... package.json	The terminal is in the wrong folder. Run <code>cd frontend</code> first.
Form says "Could not send"	Backend is not running, or frontend/.env is missing or the site was not restarted after creating it.
.env changes have no effect	The file is still named .env.example, or the server was not restarted.
Enquiry saved nowhere you look	Django may be using SQLite. Run <code>python manage.py shell -c "from django.db import connection; print(connection.vendor)"</code> . It should print oracle.
ORA-12514	Wrong service name in ORACLE_DSN, or the database or listener is stopped.
ORA-01017	Wrong Oracle user or password. Passwords are case-sensitive.
ORA-01917	The user was not created. Run <code>CREATE USER</code> alone and read its error.
ORA-01950	No tablespace quota. Run <code>ALTER USER freelance QUOTA UNLIMITED ON USERS;</code> .
ORA-00942 in SQL Developer	You are connected as a different user. Use <code>FREELANCE.ENQUIRIES_ENQUIRY</code> .
ORA-65096	You are in CDB\$ROOT. Switch to XEPDB1.
Unknown Command in SQL Developer	A statement was split across lines. Put it on one line.
Django complains about the Oracle version	This Django release needs Oracle 19c or newer(21c).
pip install fails on oracledb	Delete that line from requirements.txt and use SQLite until you need Oracle.
Browser console shows a CORS error	CORS_ORIGINS in backend/.env must match the website address exactly. Restart the backend.
Port 8000 or 5173 busy	Run <code>python manage.py runserver 8001</code> and update VITE_API_URL.
Chat says "Too many messages"	The hourly limit was reached. Wait, or raise the limit.
Chat says it is not switched on	ANTHROPIC_API_KEY is empty.
python not recognised on Windows	Try py instead.

12. Deployment outline

1. **Domain.** Buy a domain and replace `example.com` as described in section 7.
2. **Frontend.** Deploy `frontend/` to Vercel or Netlify. Build command `npm run build`, output folder `dist`. Set `VITE_API_URL` to your live backend address. The included `vercel.json` and `public/_redirects` make direct page links work.
3. **Backend.** Netlify and Vercel cannot run Django. Use a Python host such as Render, Railway or PythonAnywhere. Add a production server (gunicorn) and static-file handling, which are not included in this project yet.
4. **Database.** A local Oracle XE is not reachable from the internet. Use a hosted Oracle database (for example Oracle Cloud), which may need extra connection settings such as a wallet.
5. **Settings.** Set `DJANGO_DEBUG=0`, a strong `DJANGO_SECRET_KEY`, your real domain in `ALLOWED_HOSTS`, and your website address in `CORS_ORIGINS`.
6. **Email.** Configure real email (section 8) so enquiries reach you.

13. Security checklist

- Never commit or share `.env` files. They are already listed in `.gitignore`.
- Use a different, strong database password and secret key for production.
- Keep the API key on the server only.
- Do not post screenshots that show passwords or keys.
- The contact form has a hidden spam-trap field, server-side validation and rate limiting. Keep them when you edit the form.

14. Before you go live

- ☐ Every **Sample** badge and sample value is gone.
- ☐ Brand name, email, WhatsApp, social links and CV are real.
- ☐ Projects, pricing and any testimonials are real and verifiable.
- ☐ Domain set in `index.html`, `sitemap.xml` and `robots.txt`.
- ☐ Contact form tested end to end, and the email arrives.
- ☐ Chat assistant facts in `chat_context.py` are accurate.
- ☐ Tested on a phone and in both light and dark themes.
- ☐ Debug mode off, strong secrets in place.